3.3 Criteria for selection of the mitigation measures

The selection of the most appropriate mitigation measures to be adopted in specific situations must take into account, according to Keaton and Beckwith (1996) and SafeLand (2012), the following aspects:

- Factors which determine the hazard, in terms of the type, rate, depth and the probability of occurrence of the movement or landslide, such as, for example:
    - The physical characteristics of the geosystem, including the stratigraphy and the mechanical characteristics of the materials, the hydrological (surface water) and the hydrogeological (groundwater) regime;
    - The morphology of the area;
    - The actual or potential causative processes affecting the geosystem, which can determine the occurrence of movement or landslides;
- Factors which affect the nature and quantification of risk for a given hazard, such as the presence and vulnerability of elements at risk, both in the potentially unsta ble area and in areas which may be affected by the run-out;
- Factors which affect the actual feasibility of specific mitigation measures, such as, for example:
    - The phase and rate of movement at the time of implementation;
    - The morphology of the area in relation to accessibility and safety of workers and the public;
    - Environmental constraints, such as the impact on the archaeological, historical and visual/landscape value of the locale;
    - Pre-existing structures and infrastructure that may be affected, directly or indirectly;
    - Design standards;
    - Balancing cut and fill;
    - Capital and operating cost, including maintenance.

Note that avoiding the landside problem is an excellent approach if it is considered during the planning phase. However, a large cost may be involved if a landslide problem is detected after the location has been selected and the design completed. With that respect of avoidance, the consideration of the inventory map of the old and recent deep-seated landslides that we have identified during the first phase of the project is important. This locates the places that are the most prone to be affected by ground deformations. Avoidance should also be kept as a potential mitigation measure for the area that we have identified has the most landside hazard prone.